

Chapter 2: Prompt Engineering & AI Content Creation

Chapter-End Activity

Activity 1: Individual – Prompt Engineering Quiz & Mini Prompt Assessment

Objective

To evaluate each student's understanding of prompt engineering concepts, prompt writing techniques, AI content creation, prompt optimization, and responsible AI usage through a written quiz and practical assessment.

Materials Required

- Computer/Laptop or Tablet
 - Internet Connection
 - AI Tool (ChatGPT, Google Gemini, Microsoft Copilot, or Perplexity AI)
 - Student Quiz Worksheet
 - Mini Prompt Assessment Sheet
 - Teacher Evaluation Sheet
-

Instructions

1. Students complete the **Prompt Engineering Quiz** individually.
2. Select one topic from the list provided.
3. Write three AI prompts:
 - Research Prompt
 - Image Generation Prompt
 - Presentation Prompt
4. Optimize one prompt by adding role, context, constraints, and output format.
5. Generate AI responses and compare the results.
6. Submit the completed worksheet for evaluation.

Part A – Prompt Engineering Quiz

Multiple Choice Questions

1. What is Prompt Engineering?

- a) Writing computer programs
- b) Writing clear instructions for AI to generate accurate responses ☒
- c) Editing images manually
- d) Creating websites

2. Which component improves the quality of an AI prompt?

- a) Role and Context ☒
- b) Random keywords
- c) Short questions only
- d) Emojis

3. Which prompt is more effective?

- a) "Tell me about pollution."
- b) "Act as a Science teacher. Explain air pollution for Class X students using simple language, five key points, two real-life examples, and a conclusion." ☒

4. Why should AI-generated content be reviewed?

- a) AI never makes mistakes.
- b) To verify accuracy and improve quality. ☒

- c) It is not necessary.
 - d) To reduce internet speed.
-

5. Which is an example of an AI image prompt?

- a) "Create a 4K educational poster on Water Conservation with blue and green colors, flat illustration style, and modern typography." 
 - b) "Open Microsoft Word."
 - c) "Calculate 45×12 ."
 - d) "Increase the screen brightness."
-

Part B – Short Answer Questions

1. What is Prompt Engineering?
 2. Why are detailed prompts important?
 3. Name the four main components of an effective prompt.
 4. What is prompt optimization?
 5. What is prompt refinement?
 6. What is multi-step prompting?
 7. How can AI help create presentations?
 8. Why should AI-generated information be verified?
 9. Mention four applications of prompt engineering.
 10. Explain responsible AI usage in your own words.
-

Mini Prompt Assessment

Students should complete the following practical tasks:

- Select one academic or business topic.
- Write:
 - One Research Prompt
 - One Image Generation Prompt
 - One Presentation Prompt
- Optimize one prompt.
- Generate AI responses.

- Compare the responses.
 - Explain which prompt produced the best result and why.
-

Teacher Observation Checklist

- ✓ Writes clear and structured prompts.
 - ✓ Uses proper prompt engineering techniques.
 - ✓ Optimizes prompts effectively.
 - ✓ Reviews AI-generated responses critically.
 - ✓ Demonstrates responsible AI usage.
-

Activity 2: Team – AI Prompt Challenge & Content Creation Workshop

Objective

To develop teamwork, creativity, communication, and prompt engineering skills by creating high-quality AI-generated content for a real-world scenario.

Materials Required

- Computer/Laptop or Tablet
 - Internet Connection
 - AI Tool
 - Canva / Microsoft PowerPoint / Google Slides
 - Team Activity Worksheet
 - Teacher Evaluation Sheet
-

Instructions

1. Divide students into **groups of 3–5 members**.
 2. Each team selects one topic.
 3. Assign team roles:
 - Team Leader
 - Prompt Engineer
 - Content Creator
 - Designer
 - Presenter
 4. Each team creates:
 - One research prompt
 - One AI image prompt
 - One presentation prompt
 - One promotional content prompt
 5. Generate AI content using the prompts.
 6. Design a poster or presentation.
 7. Deliver a **3–5 minute presentation** explaining:
 - Prompt strategy
 - AI-generated content
 - Improvements made
 - Final results
 8. Participate in a question-and-answer session.
-

Suggested Team Topics

- Artificial Intelligence in Education
 - Cyber Safety Campaign
 - Smart Classroom
 - Renewable Energy
 - Digital Citizenship
 - Water Conservation
 - Healthy Lifestyle
 - Entrepreneurship
 - Space Exploration
 - Environmental Awareness
-

Team Activity Checklist

Activity	Completed (✓)
Topic Selected	☐
Research Prompt Created	☐
Image Prompt Created	☐
Presentation Prompt Created	☐
AI Content Generated	☐
Final Presentation Delivered	☐

Group Discussion Questions

1. Which prompt produced the best AI output?
 2. How did prompt refinement improve the results?
 3. What challenges did your team face?
 4. How did teamwork improve content quality?
 5. Which prompt engineering technique was most useful?
-

Teacher Observation Checklist

- ☒ Demonstrates teamwork and collaboration.
 - ☒ Creates effective prompts.
 - ☒ Produces quality AI-generated content.
 - ☒ Presents confidently.
-

Activity 3: Recognition – Best Prompt Engineer Award

Objective

To recognize students who demonstrate outstanding prompt-writing skills, creativity, logical thinking, AI communication, and responsible AI usage.

Award Criteria

Students are evaluated based on:

- Prompt Writing Skills
 - Prompt Structure
 - Prompt Optimization
 - Creativity
 - AI Response Quality
 - Critical Thinking
 - Problem Solving
 - Ethical AI Usage
 - Presentation Skills
 - Overall Performance
-

Recognition

Best Prompt Engineer Award

Awarded to the student who demonstrates exceptional skill in designing clear, detailed, and effective AI prompts that generate accurate, creative, and high-quality responses while practicing responsible AI usage.

Activity 4: Recognition – Best AI Content Creation Team Award

Objective

To recognize the team that demonstrates outstanding collaboration, creativity, prompt engineering skills, and AI-assisted content creation.

Award Criteria

Teams are evaluated on:

- Team Collaboration
- Prompt Quality
- Creativity
- AI Content Quality

- Presentation Skills
 - Communication
 - Innovation
 - Time Management
 - Equal Participation
 - Overall Performance
-

Recognition

Best AI Content Creation Team Award

Awarded to the team that demonstrates outstanding collaboration, innovative prompt engineering, excellent AI-generated content, and professional presentation skills.

Activity 5: Practical Evaluation

Objective

To assess students' overall ability to apply prompt engineering techniques for academic, creative, and professional tasks.

Practical Tasks

Students should demonstrate the ability to:

- Write effective AI prompts.
 - Generate AI responses for research and learning.
 - Create AI image prompts for posters or illustrations.
 - Generate presentation content using AI.
 - Optimize and refine prompts for improved results.
 - Evaluate AI-generated responses.
 - Demonstrate responsible AI use and ethical practices.
 - Present AI-generated work confidently.
 - Collaborate effectively in team activities.
 - Reflect on learning and identify areas for improvement.
-

Practical Evaluation Rubric

Assessment Area	Marks
Prompt Writing & Structure	10
Prompt Optimization & Refinement	10
AI Research & Content Creation	10
AI Image Prompt Design	10
Presentation Prompt Creation	10
AI Response Evaluation	10
Responsible AI Usage & Ethics	10
Creativity & Innovation	10
Presentation & Team Collaboration	10
Overall Practical Performance	10
Total	100 Marks

Teacher Observation Checklist

- ☒ Demonstrates proficiency in writing structured and effective AI prompts.
 - ☒ Uses prompt engineering techniques to generate accurate, creative, and high-quality AI content.
 - ☒ Refines prompts and evaluates AI-generated responses critically.
 - ☒ Applies responsible AI practices, ethical guidelines, and academic integrity.
 - ☒ Collaborates effectively with team members and presents AI-assisted content confidently.
-

Chapter Learning Outcomes

After completing the **Chapter-End Activities** for **Chapter 2: Prompt Engineering & AI Content Creation**, students will be able to:

- Design effective AI prompts for **research, reports, presentations, image generation, and business communication**.
- Apply advanced prompt engineering techniques, including **role-based prompting, context setting, constraints, optimization, and refinement**.
- Generate high-quality AI-assisted content for academic, creative, and professional purposes.
- Evaluate and improve AI-generated responses for **accuracy, relevance, creativity, and organization**.
- Demonstrate **responsible, ethical, and safe use of AI** while respecting academic integrity and digital citizenship.
- Collaborate effectively in teams to complete AI-powered content creation projects and confidently present their prompt strategies and outcomes.